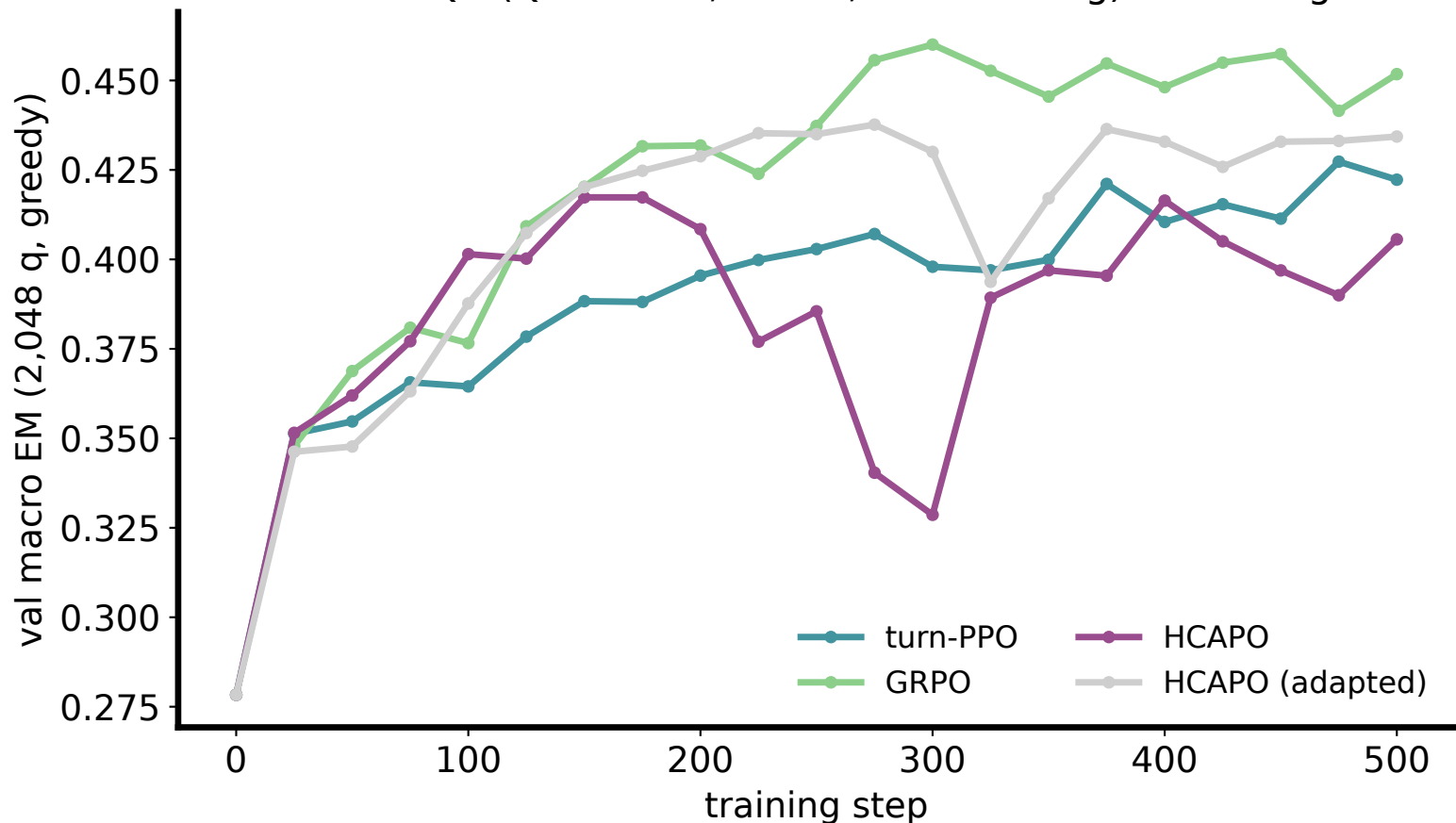


SearchQA (Qwen3-4B, 4-turn, non-thinking) — training

SearchQA (Qwen3-4B, 4-turn, non-thinking) — final eval



method	full-set macro EM	eval step
turn-PPO	0.417	500
GRPO	0.463	500
HCAPO	0.410	500
HCAPO (adapted)	0.440	500
pre-RL init	0.313	0